

Faculty Mentor Briefing Report

Project: Central Virginia Tree Canopy Policy and Benefits (UVA MSDS Capstone Project Summer 2026)

Team Members: Kyle Hoffman, Ryan Adoremos, Derrick Clarke

Executive Summary

The primary objective of this capstone initiative is to enhance environmental, economic, and social outcomes across Central Virginia by supporting evidence-based decision-making for local governments and planning organizations. The project will integrate remote sensing data (including LiDAR, satellite imagery from Soil Moisture Active Passive [SMAP], and ICESat-2 Point Clouds), administrative datasets, and statistical and predictive modeling to generate actionable insights.

The analytical framework focuses on three core areas:

1. Tree canopy and ecosystem services
2. Invasive species management
3. Policy decision-making and impact modeling

Meeting Summaries and Project Milestones

The team has conducted two formal meetings with the capstone sponsor to establish and maintain alignment on project goals, scope, and technical approaches.

Initial Kickoff Meeting (Saturday, April 25, 2026)

The primary objective of the kickoff meeting was to establish a shared understanding of project goals among all stakeholders. The team aligned on the scope of work, deliverables, and timeline for the 2026 UVA SDS Capstone Program. The session also clarified roles, responsibilities, and communication channels between the capstone team and the Central Virginia Tree Canopy organization to ensure seamless collaboration. The agenda included a review of prior work, discussion of data sources, and an overview of the five stages of team development.

Post-Meeting Action Items Completed:

- The sponsor provided the original proposal and a link to the knowledge transfer recording.
- Kyle generated GeoJSON files for the Central Virginia counties associated with the

capstone.

- Kyle created a Jupyter notebook to produce an initial set of data visualizations.
- Using instructions from UVA's IT department, Derrick installed ArcGIS Pro to access and download LiDAR data from VGIN.
- Derrick created a Python script to process the CSV file generated from ArcGIS Pro.

Second Sponsor Meeting (Monday, May 18, 2026)

The primary objective of the second meeting was to ensure continued alignment with the shared understanding of project goals, scope of work, deliverables, and timeline.

Key Updates and Discussions:

- Kyle presented on the project scope.
- Derrick provided updates on the integration and use of SMAP and LiDAR data.
- Ryan is currently creating a wireframe to support the front-end dashboard deliverable.

Analytical Framework and Technical Approach

The project employs a robust analytical framework divided into three distinct but interrelated areas of research.

1. Tree Canopy Analysis

The team is investigating the spatial distribution of tree canopy across jurisdictions, analyzing how canopy coverage has changed between 2015 and 2022 (using Charlottesville as a baseline), and identifying drivers of canopy loss or gain. Furthermore, the analysis will explore correlations between canopy coverage and land surface temperature, impervious surfaces, and socio-economic indicators.

Methodology:

The technical approach utilizes LiDAR and imagery fusion to derive canopy height and extent. The team plans to apply supervised machine learning models, such as Random Forest or CNN-based segmentation, for canopy extraction and classification. Change detection techniques will identify hotspots of canopy loss and areas of gain from mitigation or reforestation. Additionally, a Heat Island Analysis will overlay canopy coverage with land surface temperature and soil moisture levels to estimate the cooling effect of the canopy.

2. Invasive Species Analysis

This component seeks to identify where invasive species are spatially concentrated and assess their impact on native canopy health, biodiversity, and long-term sustainability.

Methodology:

The team will utilize spatial risk modeling, incorporating environmental variables such as soil type, moisture, and disturbance zones. The probability of invasion will be modeled using logistic regression and spatial machine learning models. The analysis will also focus on detecting canopy health degradation, specifically identifying canopy thinning and abnormal reflectance patterns. Data sources include the Virginia Department of Conservation & Recreation Invasive Species List and the Loudoun County Soil & Water Invasive Species Program.

3. Policy Decision-Making and Modeling

The core hypothesis of this section is that higher tree canopy coverage leads to higher K-12 test scores, lower crime rates, and improved public health outcomes.

Methodology:

The team proposes using a multivariate regression model to test this hypothesis, controlling for variables such as income, canopy density, and education. Spatial econometric models (spatial lag/spatial error models) will be employed to capture geographic dependencies. Furthermore, a Scenario Simulation Model will simulate the impacts of various policy interventions (e.g., baseline trends, increased canopy mandates, and combined canopy/cool roof policies). The team is also exploring methods for threshold detection and the development of a "Heat Island Mitigation Model."

Deliverables and Long-Term Impact

The project is structured to produce both intermediate and core outputs that will serve local governments and planning organizations.

Deliverable Type	Outputs
Intermediate Outputs	Data pipelines, cleaned datasets, exploratory analyses, and workshop summaries.
Core Outputs	Tree canopy estimates (city and counties), spatial datasets and maps, statistical analysis reports, and policy simulation models.

Long-Term Impacts:

The deliverables from this capstone project are designed to enable data-driven land-use policy, support climate resilience planning, quantify ecosystem service benefits, and

provide a foundational framework for ongoing monitoring systems in Central Virginia. Strategic recommendations based on the findings are currently under development.